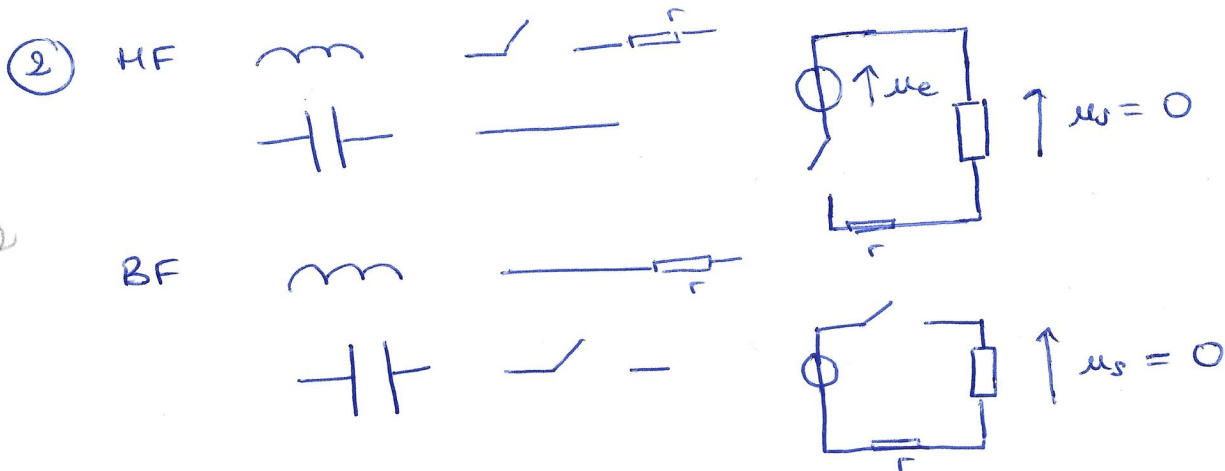
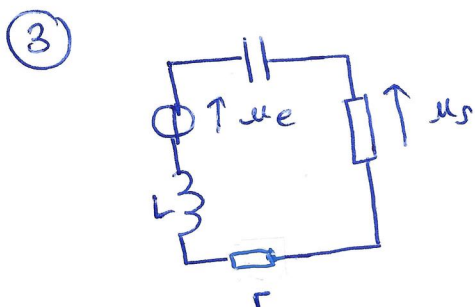


$$① \quad X_{eff} = \sqrt{\langle X(t)^2 \rangle} = \sqrt{\frac{1}{T} \int_0^T X(t)^2 dt}$$

$$② \quad \underline{u}_{eff} = \frac{E}{\sqrt{2}}$$



⇒ pass-bande



pont div de tension:

$$\underline{u}_s = \underline{u}_e \times \frac{Z_R}{Z_R + Z_C + Z_L + Z_r}$$

$$\frac{\underline{u}_s}{\underline{u}_e} = \frac{1}{1 + \frac{1}{jCR\omega} + j\frac{L\omega}{R} + \frac{r}{R}}$$

$$= \frac{R}{r + R + j\left(L\omega - \frac{1}{C\omega}\right)}$$

$$= \frac{1}{\frac{r+R}{R} + j\left(\frac{L}{R}\omega - \frac{1}{RC\omega}\right)}$$

$$\underline{H} = \frac{1}{\frac{R+r}{R} + j\left(\frac{L}{R}\omega - \frac{1}{RC\omega}\right)}$$

④

$$\underline{H} = \frac{1}{\frac{R_t}{R} + j\left(\tilde{Q}\frac{\omega}{\omega_0} - \tilde{Q}\frac{\omega_0}{\omega}\right)}$$

On pose $\omega_0 = \frac{1}{\sqrt{LC}}$

$$\tilde{Q} = \frac{1}{R}\sqrt{\frac{L}{C}}$$

(valeurs pour le RLC série vu en cours).

$$\frac{\omega_0}{\tilde{Q}} = \frac{1}{\sqrt{LC}} \cdot R\sqrt{\frac{C}{L}} = \frac{R}{L}$$

$$\tilde{Q}\omega_0 = \frac{1}{R}\sqrt{\frac{L}{C}} \cdot \frac{1}{\sqrt{LC}} = \frac{1}{RC}$$

$$\underline{H} = \frac{1}{\frac{R_t}{R} + j\tilde{Q}\left(u - \frac{1}{u}\right)}$$

$$\underline{H} = \frac{\textcircled{R/R_t} \rightarrow H_0}{1 + j \cdot \textcircled{\frac{R}{R_t} \tilde{Q}} \left(u - \frac{1}{u}\right)}$$

$\downarrow$
 Q

$$Q = \frac{R}{R_t} \cdot \frac{1}{R} \sqrt{\frac{L}{C}}$$

ici: $Q = \frac{1}{R_t} \sqrt{\frac{L}{C}}$

$$H_0 = \frac{R}{R_t}$$

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

$$u = \frac{\omega}{\omega_0}$$

3

$$\underline{H} = \frac{H_0}{1 + jQ\left(u - \frac{1}{u}\right)}$$

⑤ $\omega = \omega_c$ tel que

$$G(\omega_c) = \frac{G_{\max}}{\sqrt{2}}$$

$$G_{\max} = H_0.$$

$$⑥ \quad G = \frac{G_{\max}}{\sqrt{2}} \quad \Leftrightarrow \quad \frac{H_0}{\sqrt{2}} = \frac{H_0}{\sqrt{1 + Q^2(u - 1/u)^2}}$$

$$\Leftrightarrow 2 = 1 + Q^2(u - 1/u)^2$$

$$\Leftrightarrow 1 = Q^2(u - 1/u)^2$$

$$\Leftrightarrow Q(u - 1/u) = 1 \text{ ou } Q(u - 1/u) = -1$$

$$\Leftrightarrow u - \frac{1}{u} - \frac{1}{Q} = 0 \text{ ou } u - \frac{1}{u} + \frac{1}{Q} = 0$$

$$\Leftrightarrow u^2 - \frac{u}{Q} - 1 = 0 \text{ ou } u^2 + \frac{u}{Q} - 1 = 0$$

$$\Delta = \left(\frac{1}{Q}\right)^2 + 4 \quad u = \left(\frac{1}{Q} \pm \sqrt{\Delta}\right) \cdot \frac{1}{2} \quad u = \left(-\frac{1}{Q} \pm \sqrt{\Delta}\right) \frac{1}{2}$$

$$u > 0 \text{ donc } u = \left(\frac{1}{Q} + \sqrt{\Delta}\right) \frac{1}{2} \text{ ou } u = \left(-\frac{1}{Q} + \sqrt{\Delta}\right) \frac{1}{2}$$

$$\omega_1 = \frac{\omega_0}{2} \left(\frac{1}{Q} + \sqrt{\frac{1}{Q^2} + 4} \right)$$

$$\omega_2 = \frac{\omega_0}{2} \left(-\frac{1}{Q} + \sqrt{\frac{1}{Q^2} + 4} \right)$$

$$⑦ \quad \omega_1 - \omega_2 = \frac{\omega_0}{2} \left[\frac{1}{Q} + \frac{1}{Q} \right] = \frac{\omega_0}{2} \cdot \frac{2}{Q} = \frac{\omega_0}{Q}$$

$$\wedge \quad \Delta\omega = \frac{\omega_0}{Q}$$

⑧

A HF

$\omega \rightarrow +\infty$

$$\underline{H} \sim \frac{H_0}{j\omega} \sim -j \frac{H_0}{\omega}$$

$$G = |\underline{H}| \sim \frac{H_0}{\omega}$$

$$G_{dB} = 20 \log G \sim 20 \log \left(\frac{H_0}{\omega} \right) \\ = 20 \log \frac{H_0}{\omega} - 20 \log \omega$$

asymptote -20 dB/dec .

$$\varphi \rightarrow -\pi/2$$

A BF

$\omega \rightarrow 0$

$$\underline{H} \sim \frac{H_0}{-j\omega} \sim j \frac{H_0}{\omega}$$

$$|G| = |\underline{H}| = \frac{H_0}{\omega}$$

$$G_{dB} = 20 \log G \sim 20 \log \frac{H_0}{\omega} + 20 \log \omega$$

asymptote $+20 \text{ dB/dec}$

$$\varphi \rightarrow +\pi/2$$

⑨

Correspond au graphe

⑩

$$G_{dB}(\omega_c) = G_{dBmax} - 3 \text{ dB}$$

⑪

$$G_{dBmax} = -6 \text{ dB}$$

$$\Delta f \sim 1 \cdot 10^7 \text{ Hz}$$

⑫

$$f_0 \sim 5 \cdot 10^8 \text{ Hz}$$

$$\Delta \omega = \frac{\omega_0}{Q}$$

$$\Rightarrow Q = \frac{\omega_0}{\Delta \omega} = \frac{2\pi \cdot f_0}{2\pi \Delta f} = \frac{f_0}{\Delta f}$$

$$= \frac{5 \cdot 10^8}{10^7} \approx \underline{\underline{50}} \\ \approx \underline{\underline{5 \cdot 10^1}}$$

$$\omega_0 = 2\pi f_0 = 2\pi \cdot 5 \cdot 10^8 \\ \approx 3.1 \cdot 10^9 \sim \underline{\underline{3 \cdot 10^9 \text{ rad/s}}}$$